



**Digital cellular telecommunications system (Phase 2+) (GSM);
Universal Mobile Telecommunications System (UMTS);
LTE;
5G;
Alphabets and language-specific information
(3GPP TS 23.038 version 19.0.0 Release 19)**



6 Individual parameters

6.1 General principles

6.1.1 General notes

Except where otherwise indicated, the following shall apply to all character sets:

- 1: The characters marked "1)" are not used but are displayed as a space.
- 2: The characters of this set, when displayed, should approximate to the appearance of the relevant characters specified in ISO 1073 [16] and the relevant national standards.
- 3: Control characters:

Code	Meaning
LF	Line feed: Any characters following LF which are to be displayed shall be presented as the next line of the message, commencing with the first character position.
CR	Carriage return: Any characters following CR which are to be displayed shall be presented as the current line of the message, commencing with the first character position.
SP	Space character.

- 4: The display of characters within a message is achieved by taking each character in turn and placing it in the next available space from left to right and top to bottom.

6.1.2 Character packing

6.1.2.1 SMS Packing

6.1.2.1.1 Packing of 7-bit characters

If a character number α is noted in the following way:

b7 b6 b5 b4 b3 b2 b1
 $\alpha a \alpha b \alpha c \alpha d \alpha e \alpha f \alpha g$

The packing of the 7-bit characters in octets is done by completing the octets with zeros on the left.

For examples, packing: α

- one character in one octet:

- bits number:

7 6 5 4 3 2 1 0
 0 1a 1b 1c 1d 1e 1f 1g

- two characters in two octets:

- bits number:

7 6 5 4 3 2 1 0
 2g 1a 1b 1c 1d 1e 1f 1g
 0 0 2a 2b 2c 2d 2e 2f

6.2 Character sets and coding

This section provides list of character sets and codings to be supported by SMS, CBS, USSD and IEs included in NAS messages as specified in 3GPP TS 24.008 [20] and 3GPP TS 24.301 [21]. Implementation of the GSM 7 bit default alphabet is mandatory. Support of other character sets is optional.

It should be noted that support of Latin and non-Latin languages by GSM 7 bit default alphabet is limited. It is therefore essential to introduce UCS 2 character set in mobile stations, SCs and systems handling SMSs, CBSs, USSDs, and IEs included in NAS messages.

6.2.1 GSM 7 bit Default Alphabet

Bits per character: 7

CBS/USSD/IE of NAS message pad character: CR

Character table:

				b7	0	0	0	0	1	1	1	1
				b6	0	0	1	1	0	0	1	1
				b5	0	1	0	1	0	1	0	1
b4	b3	b2	b1		0	1	2	3	4	5	6	7
0	0	0	0	0	@	Δ	SP	0	i	P	¿	p
0	0	0	1	1	£	_	!	1	A	Q	a	q
0	0	1	0	2	\$	Φ	"	2	B	R	b	r
0	0	1	1	3	¥	Γ	#	3	C	S	c	s
0	1	0	0	4	è	Λ	α	4	D	T	d	t
0	1	0	1	5	é	Ω	%	5	E	U	e	u
0	1	1	0	6	ù	Π	&	6	F	V	f	v
0	1	1	1	7	ì	Ψ	'	7	G	W	g	w
1	0	0	0	8	ò	Σ	(	8	H	X	h	x
1	0	0	1	9	Ç	Θ	)	9	I	Y	i	y
1	0	1	0	10	LF	Ξ	*	:	J	Z	j	z
1	0	1	1	11	Ø	1)	+	;	K	Ä	k	ä
1	1	0	0	12	ø	Æ	,	<	L	Ö	l	ö
1	1	0	1	13	CR	æ	-	=	M	Ñ	m	ñ
1	1	1	0	14	Å	ß	.	>	N	Ü	n	ü
1	1	1	1	15	å	É	/	?	O	§	o	à

NOTE 1): This code is an escape to an extension of this table (either to the GSM 7 bit default alphabet extension table, see subclause 6.2.1.1, or a National Language Single Shift Table, see subclause 6.2.1.2.2). A receiving entity which does not understand the meaning of this escape mechanism shall display it as a space character.

6.2.1.1 GSM 7 bit default alphabet extension table

The table below is reserved for symbols of international significance (e.g currency symbols). It also contains a mechanism to permit escape (Note 1) to additional tables for symbols of international significance in the event that the table below becomes fully populated.

				b7	0	0	0	0	1	1	1	1
				b6	0	0	1	1	0	0	1	1
				b5	0	1	0	1	0	1	0	1
b4	b3	b2	b1		0	1	2	3	4	5	6	7
0	0	0	0	0								
0	0	0	1	1								
0	0	1	0	2								
0	0	1	1	3								
0	1	0	0	4		^						
0	1	0	1	5							€	
0	1	1	0	6								
0	1	1	1	7								
1	0	0	0	8			{					
1	0	0	1	9			}					
1	0	1	0	10	3)							
1	0	1	1	11		1)						
1	1	0	0	12			[					
1	1	0	1	13			~					
1	1	1	0	14			]					
1	1	1	1	15			\					

In the event that an MS receives a code where a symbol is not represented in the above table then the MS shall display either the character shown in the main GSM 7 bit default alphabet table in subclause 6.2.1., or the character from the National Language Locking Shift Table in the case where the locking shift mechanism as defined in subclause 6.2.1.2.3 is used.

NOTE 1): This code is reserved for the extension to another extension table. On receipt of this code, a receiving entity shall display a space until another extension table is defined. It is not intended that this extension mechanism should be used as an alternative to UCS2 to enhance the 7bit default alphabet character repertoire for national specific character sets.

NOTE 2): Void

NOTE 3): This code is defined as a Page Break character and may be used for example in compressed CBS messages. Any mobile station which does not understand the GSM 7 bit default alphabet table extension mechanism will treat this character as Line Feed.